

PHYSICAL PROPERTIES

Part 2

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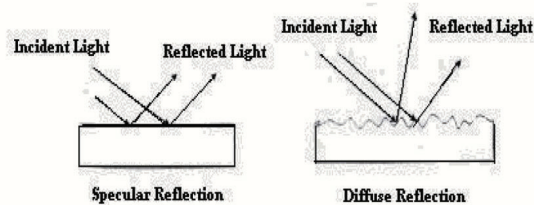
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Interaction between light and matter:

When a beam of light falls on a surface of a medium, the following may occur:

1. Reflection:

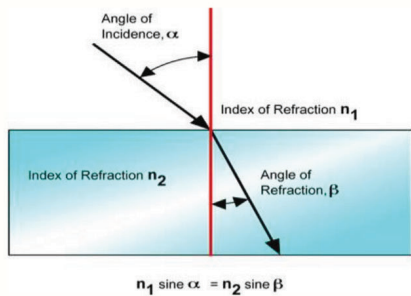
- When light falls on a **smooth surface** (perfectly smooth), the angle of incidence will be equal to the angle of reflectance.
- i.e. Smooth **surfaces reflect** light in one direction only and this is called spectral reflection. Such surface appears **shiny**.
- If light falls on a **rough surface** it will be reflected in all directions, and this is called **diffuse reflection**, Rough surfaces appear **dull**.



Reflection of light over smooth and rough surfaces

2. Refraction:

- It is the change of the direction of a beam of light on entering second medium.
- Refraction results from the difference in refractive indices of the two media.

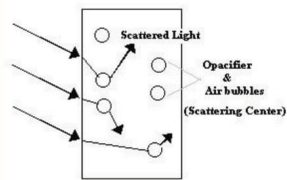


Refractive index

3. Scattering:

- If light rays passing through a medium are obstructed by any different inclusions it will be redirected in another direction and attenuated.

The original beam is weakened by scattering in a direction away from the observer eye.



Scattering centers

4. Transmission (passage of light):

- Total transmission occurs in perfectly transparent material.
- If part of the light is **transmitted** and part is reflected (i.e. diffuse transmission), the material appears **translucent**.

5. Absorption:

- As light passes through an optical medium, the electrons in the material may absorb the electromagnetic energy of that light.
- The more **absorption** of light, the more **opaque** is the material.

Properties of materials in relation to light transmission:

i. Transparency:

- A property of the material, that allows the passage of light so that an object **can be clearly** seen through it (e.g. glass and acrylic resin).

ii. Translucency:

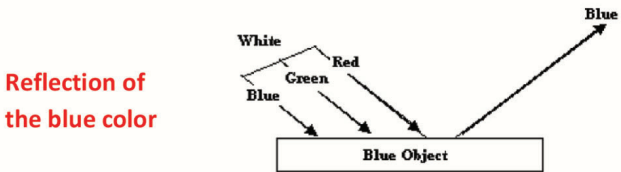
- A property of the material that allows the passage of some light and scatters or reflects the rest.
- The object **cannot be clearly** seen through it. (e.g. tooth enamel, porcelain, composite and pigmented acrylic resin.).

NB

Translucency decreases with increasing the scattering centers.

iii. Opacity:

- A property of the material that prevents the passage of light.
- Objects **cannot be seen** through it.
- Black color materials absorb all light colors.
- White color materials reflect all light colors.
- Blue color materials absorb all light colors but reflect their own color.



Reflection of the blue color

6. Luminescence (Fluorescence and Phosphorescence):

- It is a reemission of light of longer wavelength after absorption of energy of shorter wavelength from various sources. (i.e. after activation by light)
- **Immediate emission** is called fluorescence.
- **Delayed emission** is called phosphorescence.

Color

Color parameters:

i) Hue:

- It is the dominant wavelength. (Name of color)

ii) Chroma:

- It represents the strength (concentration) of color or degree of saturation of the color.
- A color may be of low chroma (0) or high chroma (10).

iii) Value:

- It represents the lightness or darkness of color (the amount of grayness).
- Value is the most important parameter of color in dentistry **because it is related to the vitality of human teeth**.
- A black standard have 0 value, whereas a white standard have 100 value

Primary colors:

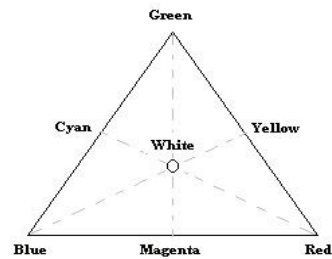
Blue, green and red are three primary colors. Combining suitable portion of wavelengths of three primary colors will produce white color

Secondary colors:

Combination of two primary colors will give the secondary colors: **yellow, cyan and magenta**.

Complementary colors:

- Two colors (**one primary+ one secondary**) are complementary to each other when their combination gives rise to white color.
- **Yellow** is the complementary of **blue**
- **magenta** is the complementary of **green**
- **cyan** is the complementary of **red**.



Primary and secondary colors

Factors affecting color appearance and selection:

"**Shade guide** is used for color matching. Therefore, it is important to match colors under appropriate conditions."

1. Source:

- **Metamerism**

Objects that appear to be color matched under one type of light may appear different under another light source.

- Thus, if possible, color matching should be done under **daylight**.
- Daylight contains light of almost all visible wavelengths, whereas most electric lamps emit most of their light over a limited wavelength range.

2. Surrounding:

- Colors of wall, lips or clothes of the patient modify the type of light reaching the object.

3. Object

a. Translucency:

- It controls the lightness or darkness of color. High translucency gives a lighter color appearance (higher value) i.e. more vital tooth appearance.

b. Surface texture (surface finish):

- This determines the relative amount of light reflected from the surface.
- Smooth surface appear brighter than rough surface.

c. Presence of scattering centers

- As inclusions or voids, which increase opacity and lower the value (more dark).

d. Fluorescence:

- It makes the human teeth bright and vital, as it increases the brightness.

e. Thickness:

- The thickness of a restoration can affect its appearance.

f. Metamerism:

i- Metameric pair:

- Two objects that are matched in color under one light source but are not matched under other light sources form metameric pair.

ii- Isomeric pair:

- They are color matched under all light sources.

4. Observer:

a. Color response:

- Eye responds differently among individuals.

b. Color vision:

- Some individuals may have color blindness and inability to distinguish certain colors.

c. Color fatigue:

- Constant stimulus of one color decreases the response to that color.
- In order to solve the problem of color fatigue, that occur by looking too long at the tooth (yellow color), the eye should be subjected to look at intervals to the complementary color (blue color).

IV. Other properties:

1. Water sorption:

- It represents the amount of water **adsorbed** on the surface and **absorbed** into the body of the material.

2. Fluidity, Viscosity and Plasticity:

- **Fluidity** is the tendency of liquids to flow.
- **Viscosity** is the resistance of the liquid to flow.
- **Plasticity** is a property related to solids or semisolids and indicates that the material is easily and permanently deformed under force.

3. Shelf Life:

- It is a term applied to the general deterioration and change in quality of materials during shipment and storage.

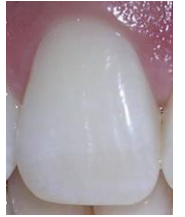
Physical properties of Non-metallic esthetic restorative materials:

- 1. Reflection:** The restoration should have a highly smooth and polished surface to reflect light in one

direction to appear shiny to simulate the tooth structure and match its color.

2. Refraction: Refractive index of the tooth and the esthetic restorative material should match each other, to avoid **line of demarcation** .

Line of demarcation



3. Scattering: Presence of scattering centers as opacifiers and air bubbles in an esthetic restoration causes light to emerge in all directions, leading to increase opacity.

e.g.:

a) **Opacifiers** added to composite resins act as scattering centers that give rise to more opaque shades of the materials.

b) Incorporation of **air bubbles** in the restoration during mixing acts as **scattering centers**

4. Fluorescence: anterior restorative materials and dental porcelains are formulated with fluorescing agents to reproduce the natural appearance of the tooth structure.

as sound human teeth emit fluorescent light when excited by ultraviolet radiation.

This fluorescing contributes to the **brightness** and vital appearance of human teeth.

5. High translucency and High value: High translucency gives a lighter color appearance (higher value) . more vital tooth appearance. Increase in thickness will increase opacity(lower value).

6. Should form isomeric pair with tooth structure: color matched under all light sources